Final Report - Medium-Term Ticker Prediction Using Options Data and Machine Learning Models

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1. Executive Summary

Background: Predicting stock price movements remains one of the most complex challenges in financial research due to the multitude of variables influencing price movements and the market's unpredictable nature. The underlying assumption of our project is that options trading data reflects informed traders' bets on tickers' future price movements and can offer some insights that we can attempt to capture and utilize. In this capstone project, we developed a data pipeline and machine learning models to leverage options data for medium-term stock price movement predictions. Our evaluation was based on simulations utilizing these predictions.

Objective:

- Build a robust data pipeline to collect, process, and aggregate trading options data for the S&P 500.
- Design machine learning models to predict stock price movements over a two-month horizon.
- Run simulations to evaluate the profitability of model-based investment strategies compared to the market.

Methods:

- **Data Pipeline**:
    - Gathered options data (100,000+ contracts, 20+ million data points) using the Polygon API.
    - Engineered features, including Greeks (Delta, Gamma, Theta, Vega, Rho) and implied volatility via the Black-Scholes model.
    - Aggregated data by moneyness and calculated momentum indicators like MACD.
- **Machine Learning**:
    - Ran multiple models (e.g. XGBoost, LSTM) with and without PCA for dimensionality reduction.
- **Evaluation**:
    - Simulated an investment strategy based on model predictions to evaluate the ROI.

Results: A fully functional data pipeline and machine learning framework were developed to predict medium-term stock movements. The models achieved around 54% accuracy, and the investment strategy based on our best-performing models' predictions outperformed the S&P 500 ROI in the majority of simulation months.

Conclusion: This capstone demonstrated a promising approach for leveraging options data to predict stock price movements and guide investment strategies. Future work includes expanding our dataset's time range, building a platform for users to view insights, and experimenting with additional ML models.

2. Problem Definition and Goal

Stock market forecasting is one of the most challenging problems in financial research. Predicting whether a stock will go up or down is difficult due to the number of variables influencing stock prices, from macroeconomic factors to company-specific events. Even small changes can lead to major market shifts, making the stock market notoriously unpredictable. As a result, many complex models, ranging from traditional statistical techniques to cutting-edge machine learning models, have been developed over the years to predict stock prices. However, despite their sophistication, many of these models have fallen short in accurately forecasting stock movement, particularly in the short to medium term.

For this project in collaboration with JPMorgan, we've decided to focus on options data as a key source of predictive information. The logic behind this is that options essentially represent market participants placing bets on where they think a stock's price will move in the future. By analyzing how options are priced, we can gather insights into the collective expectations of traders and investors. This approach could offer an additional layer of predictive power compared to traditional stock price data alone.

2.1. Background Information

A central tool in our approach is the Black-Scholes model, which is widely used for calculating the theoretical price of options. This model uses factors like the current stock price, the strike price of the option, the time until the option expires, the risk-free interest rate, and the stock's volatility to estimate what an option should be worth. While the model simplifies some market dynamics, it remains one of the core models for options pricing.

Key to the Black-Scholes model are the **Greeks**, which measure the sensitivity of the option's price to different factors. These include:

- **Delta**: Measures the sensitivity of the option's price to changes in the underlying stock price.

$$\Delta_{call} = N(d_1)$$

$$\Delta_{put} = N(d_1) - 1$$

- **Gamma**: Represents the rate of change of delta concerning the stock price.

$$\Gamma = \frac{N'(d_1)}{S_0 \sigma \sqrt{T}}$$

- **Theta**: Indicates how the option's price changes as it approaches its expiration date, measuring time decay.

$$\Theta_{call} = -\frac{S_0 N'(d_1)\sigma}{2\sqrt{T}} - rXe^{-rT}N(d_2)$$

$$\Theta_{put} = -\frac{S_0 N'(d_1)\sigma}{2\sqrt{T}} + rXe^{-rT}N(-d_2)$$

- **Vega**: Reflects the sensitivity of the option's price to changes in volatility of the underlying stock.

$$\nu = S_0 N'(d_1)\sqrt{T}$$

- **Rho**: Shows how the option price reacts to changes in interest rates.

$$\rho_{call} = XTe^{-rT}N(d_2)$$

$$\rho_{put} = -XTe^{-rT}N(-d_2)$$

- **Implied Volatility**: The market's expectation of future stock price fluctuations, inferred from the option's price. It reflects how much traders anticipate the stock will move, regardless of direction.
    - Calculated from the market price of the option, using the Black-Scholes formula, by solving for σ, the implied volatility.

For more details on the terms used, please refer to the Appendix.

By incorporating these Greeks as features in our forecasting model, we aim to capture valuable information from the options market. This should give us a better sense of market sentiment and help us improve our predictions on whether a stock will move up or down in the next two months.

2.2. Related Works

Forecasting stock market trends has always been a challenging task due to the complexity and volatility of financial markets. Traditional models like the Black-Scholes formula have allowed us to predict options pricing by offering a mathematical framework to estimate option premiums based on variables such as stock price, volatility, and time to expiration. However, these models rely on constant volatility and efficient markets, which often do not hold in practice (Yuan & Chen, 2024; Chowdhury et al., 2020). This has led researchers to explore additional approaches that incorporate more dynamic data sources and modified methodologies to improve forecasting accuracy.

2.3. Challenges with Traditional Models

The Black-Scholes model has been widely used for pricing European-style options, but it has limitations in the real world. Chowdhury et al. (2020) point out that factors like volatility and risk-free interest rates fluctuate, creating gaps between theoretical prices and actual market outcomes. To address these issues in frontier markets, their study adjusted strike prices and expiration periods to reflect less frequent trading and irregular contract purchases, making the model more adaptable to those environments.

Other traditional models, such as ARIMA and elastic net regression, have also been applied to financial forecasting. ARIMA is useful for capturing trends in historical data but struggles with sudden market shifts. Similarly, elastic net, which combines LASSO and ridge regression, helps manage multicollinearity but is limited in dealing with non-linear behaviors and market volatility. These models, while effective in certain scenarios, often require complementary approaches to handle the complexities of financial markets (Song et al., 2024; Yuan & Chen, 2024).

2.4. The Efficient Market Hypothesis

One highly relevant concept that must be addressed in this context is *the Efficient Market Hypothesis (EMH),* one of the most prominent theories in financial research. According to Mehwish Naseer and Yasir bin Tariq, this hypothesis states that investors can't make predictions or earn above-average risk-adjusted profit using new information as it is quickly incorporated into security prices and market activities. This implies that markets are highly efficient and leave little room for investors to consistently outperform the market by analyzing historical or current data (Mehwish & bin Tarik, 2016). This hypothesis can be categorized into three forms: weak, semi-strong, and strong. First, weak form EMH implies that a security's price reflects all historical market information making any market analysis ineffective. Second, semi-strong EMH suggests that securities prices adjust quickly to publicly available information such as dividend and earning announcements, rendering any fundamental analysis unproductive. Finally, strong-form EMH proposes that securities prices reflect the entirety of public and private information regarding those securities and that no investor has monopolistic access to information (Mehwish & bin Tarik, 2016).

The Efficient Market Hypothesis has garnered critical attention since its introduction and many of its shortcomings have been discussed in the literature. According to Andrew Lo, some investor traits such as overconfidence, herd mentality, and loss aversion introduce inefficiencies that go against this hypothesis (Lo, 2007). Additionally, anomalies such as the January effect and the Value Line enigma pose serious challenges to the EMH. As Lo stated: "... [the anomalies'] persistence in the face of public scrutiny seems to be a clear violation of the EMH" (Lo, 2007), and he further noted: "most of these anomalies can be exploited by relatively simple trading strategies" (Lo, 2007). Finally, Grossman and Stiglitz argue that perfectly efficient markets are an impossibility since eliminating incentives relating to gathering information would result in a diminishing of trade, and consequently, a market collapse (Grossman and Stiglitz, 1980).

The efficient market hypothesis poses a major challenge concerning this capstone project. However, its shortcomings introduce opportunities that we hope our predictive models will capitalize on. Specifically, by analyzing options trading data and relevant options metrics and indicators, we aim to capture market sentiment that may not be solely reflected in stock prices, identify some inefficiencies, and leverage them in a simulated investing strategy.

2.5. Incorporating Market Dynamics

Studies show that integrating more dynamic aspects of financial markets can improve the accuracy of forecasting models. Chia (2016) explored the predictive power of options market data, such as implied volatility and put-call ratios, to gain insights into stock price movements. This work emphasizes that options data reflects the collective expectations of market participants, providing valuable information not captured by traditional stock price data alone. Yuan and Chen (2024) further built on this concept by combining historical price data with real-time sentiment indicators to effectively adapt to shifting market conditions.

2.6. Hybrid Approaches to Forecasting

Given the limitations of using a single forecasting model, hybrid approaches have been tested to improve accuracy. Song et al. (2024) proposed a hybrid framework combining nonparametric regression techniques with Long Short-Term Memory (LSTM) models to forecast stock prices. Their method captures both long-term trends and short-term fluctuations by first using the non-parametric modeling to capture the overall trend with kernel functions, then using an LSTM framework to predict the residuals for the periods and lastly using linear reconstruction to combine the output of the non-parametric regression and LSTM to have a more refined prediction. Similarly, Chowdhury et al. (2020) demonstrated how a modified Black-Scholes framework, paired with machine learning techniques, can enhance stock price forecasting by dynamically adjusting inputs to reflect changing market conditions.

2.7. Using Options Data to Predict Stock Movement

Building on these approaches, integrating options market data as predictive features offers significant potential to enhance stock price forecasting. Options data - including implied volatility, open interest, and put-call ratios—reflects market sentiment and traders' expectations of future price movements. When incorporated into forecasting models, can improve predictions by capturing both market trends and investor behavior. Chia (2016) showed that treating option contracts as indicators of future stock movements adds predictive value beyond traditional time-series models. Combining these features with modified Black-Scholes parameters, such as volatility and strike price adjustments, creates more powerful forecasting models suited for complex market conditions.

By leveraging the predictive power of options data alongside hybrid modeling techniques, forecasting tools can achieve higher accuracy. This combined approach makes it possible to account for both technical market indicators and behavioral sentiment, helping traders and analysts make better-informed decisions. Incorporating options data as features in forecasting models represents a promising direction for future research, offering a more comprehensive view of market dynamics and improving the ability to predict stock movement.

3. EDA

For the Exploratory Data Analysis (EDA), we chose to focus our analysis on the Apple stock. This EDA can be easily replicated for all of the S&P 500 stocks as we have extracted that data as well. Figure 1's chart looks at the relationship between AAPL's stock price and implied volatility to understand market sentiment and potential risk. From January 2023 to mid-2024, AAPL's stock price steadily climbed while implied volatility stayed low, suggesting that investors weren't expecting major price swings. But in early 2024, implied volatility spiked sharply, even though the stock price didn't fluctuate much. This likely signals a shift in market expectations, possibly in response to upcoming events like a product launch. The key takeaway is that a rise in implied volatility, even when the stock price remains stable, could indicate that the market is preparing for potential volatility, offering insight into future risks or opportunities with AAPL.

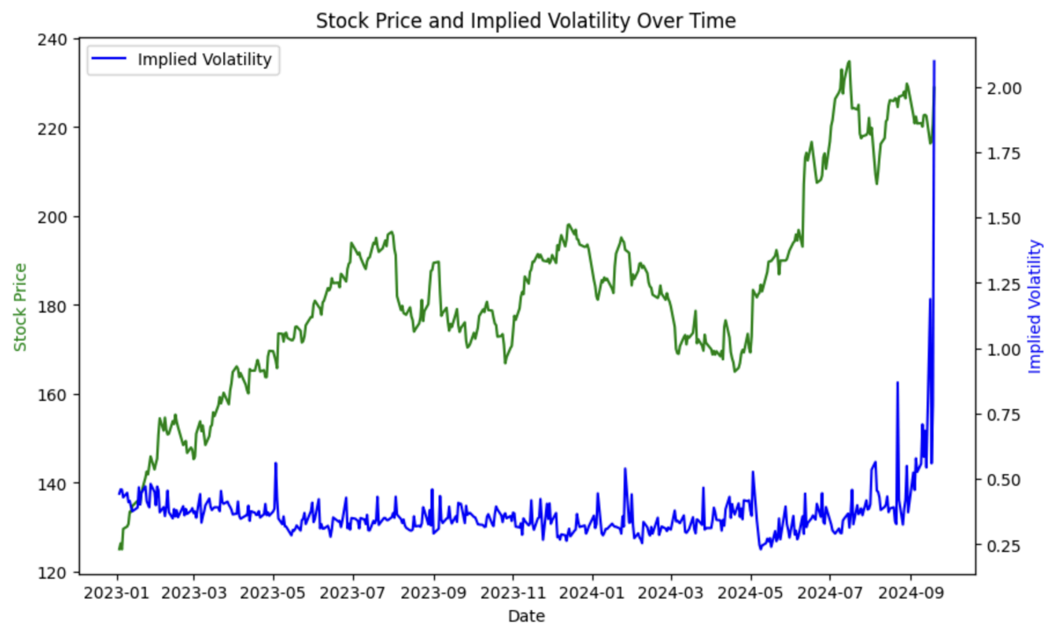

Figure 4: AAPL Stock Price vs. Implied Volatility Over Time, Highlighting Market Sentiment and Risk

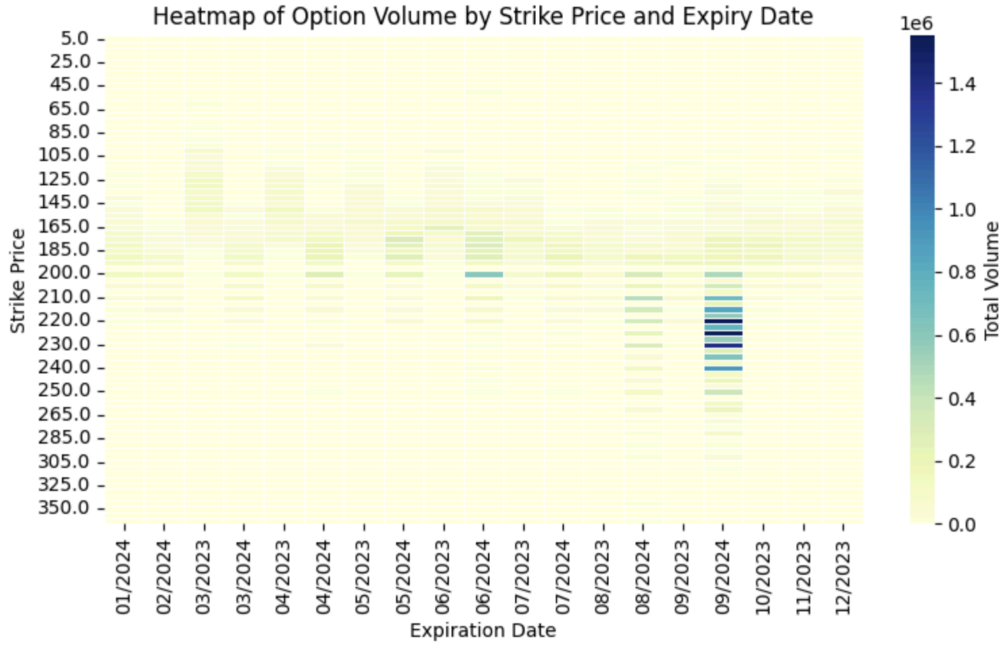

Figure 2: Heatmap of Option Volume by Strike Price and Expiration Date, Highlighting Investor Focus and Anticipated Price Levels

This heat map illustrates the distribution of option volume by **strike price** and **expiration date**. The darker regions indicate higher volumes, with a noticeable concentration of activity around strike prices between 200 and 240, especially toward the end of 2024. This suggests that investors are focusing their option trades in this price range, possibly expecting AAPL's stock to fluctuate around these levels. The increase in volume around later expiration dates could indicate growing interest in longer-term options as market participants position themselves for potential movements in AAPL's price. The activity clustered around September 2024 might reflect anticipation of significant market events around that time.

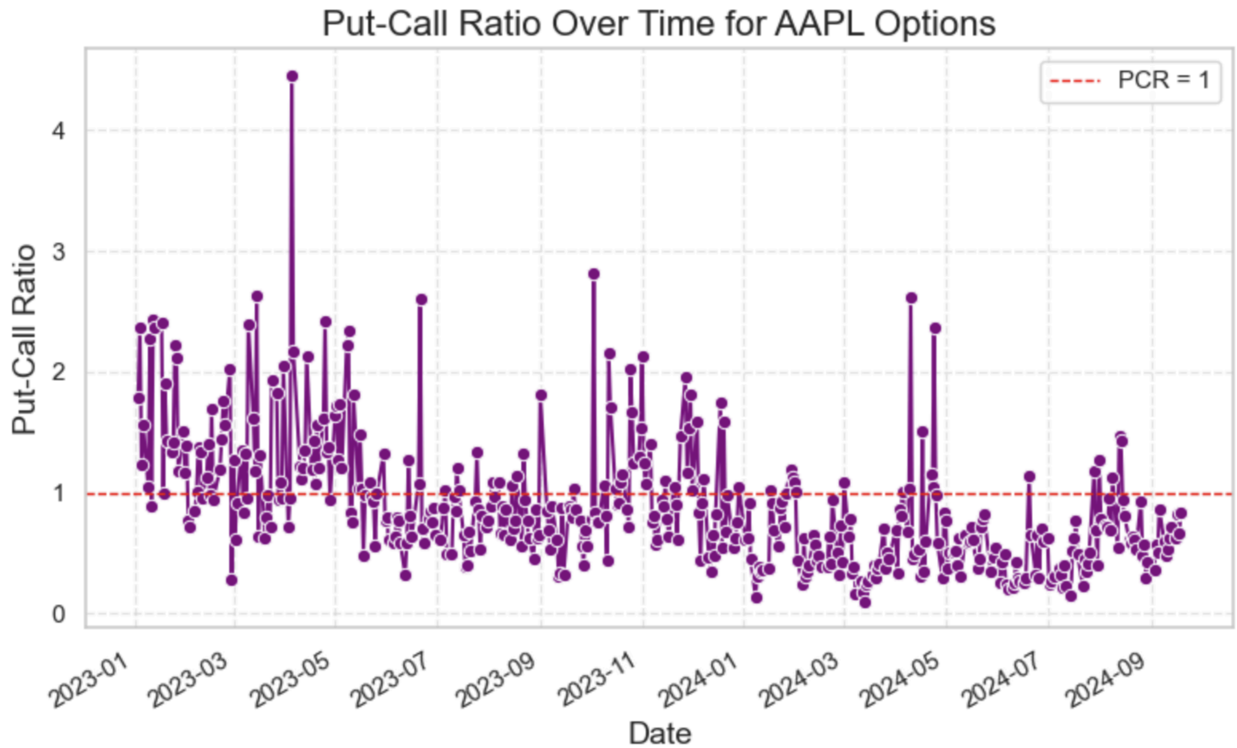

Figure 3: Put-Call Ratio Over Time for AAPL Options, Indicating Shifts in Market Sentiment

This chart shows the **Put-Call Ratio (PCR)** for AAPL options over time, providing insight into market sentiment. The PCR measures the number of put options traded relative to call options, with a ratio above 1 indicating more puts (bearish sentiment) and below 1 indicating more calls (bullish sentiment). Throughout 2023, the PCR fluctuated significantly, with several spikes above 3, indicating a strong bearish sentiment during these periods. As time progresses into 2024, the PCR stabilizes closer to 1, suggesting a more balanced or neutral sentiment between puts and calls. However, there are still occasional spikes, signaling brief periods of increased bearish activity. The red dashed line at **PCR = 1** acts as a benchmark, showing where the market sentiment shifts between bullish and bearish outlooks. Overall, this chart helps identify key moments of heightened concern or optimism among AAPL investors, with a trend toward more neutral sentiment as 2024 unfolds.

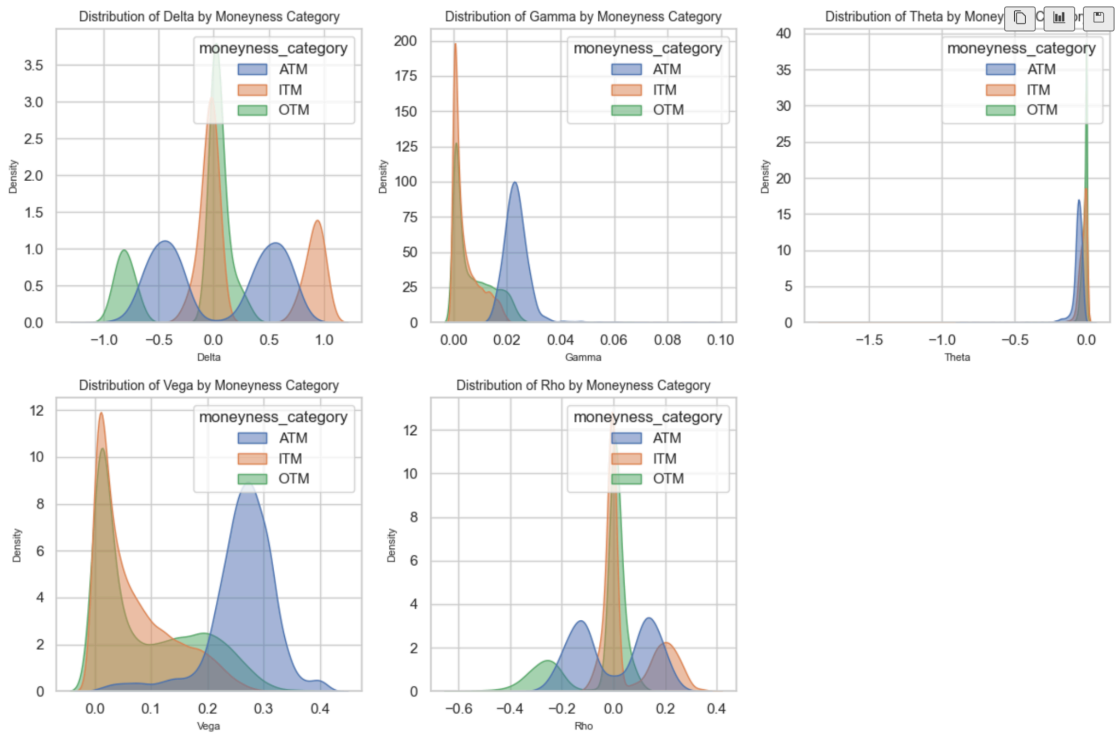

Figure 4: Distribution of Greeks by Moneyness Category (ATM, ITM, OTM)

This plot matrix shows the distributions of five key Greeks — **Delta**, **Gamma**, **Theta**, **Vega**, and **Rho** — categorized by **moneyness** (ATM, ITM, and OTM).

- **Delta**: The distribution reflects the expected behavior based on money. In-the-money (ITM) options have deltas around +1/0, out-of-the-money (OTM) options are centered near 0, and at-the-money (ATM) options cluster around the midpoint.
- **Gamma**: ATM options have the highest gamma, showing the greatest sensitivity to stock price changes, while ITM and OTM options show lower gamma, which is typical as they have less sensitivity.
- **Theta**: All options have negative theta values, indicating time decay, with ITM and ATM options experiencing more time decay than OTM options. This reflects the fact that time decay accelerates as options approach expiration, particularly for ITM and ATM contracts.
- **Vega**: ATM options have the highest vega, meaning they are most sensitive to changes in volatility, while OTM and ITM options show less sensitivity.
- **Rho**: ITM options show the most significant response to interest rate changes, while ATM and OTM options exhibit less sensitivity.

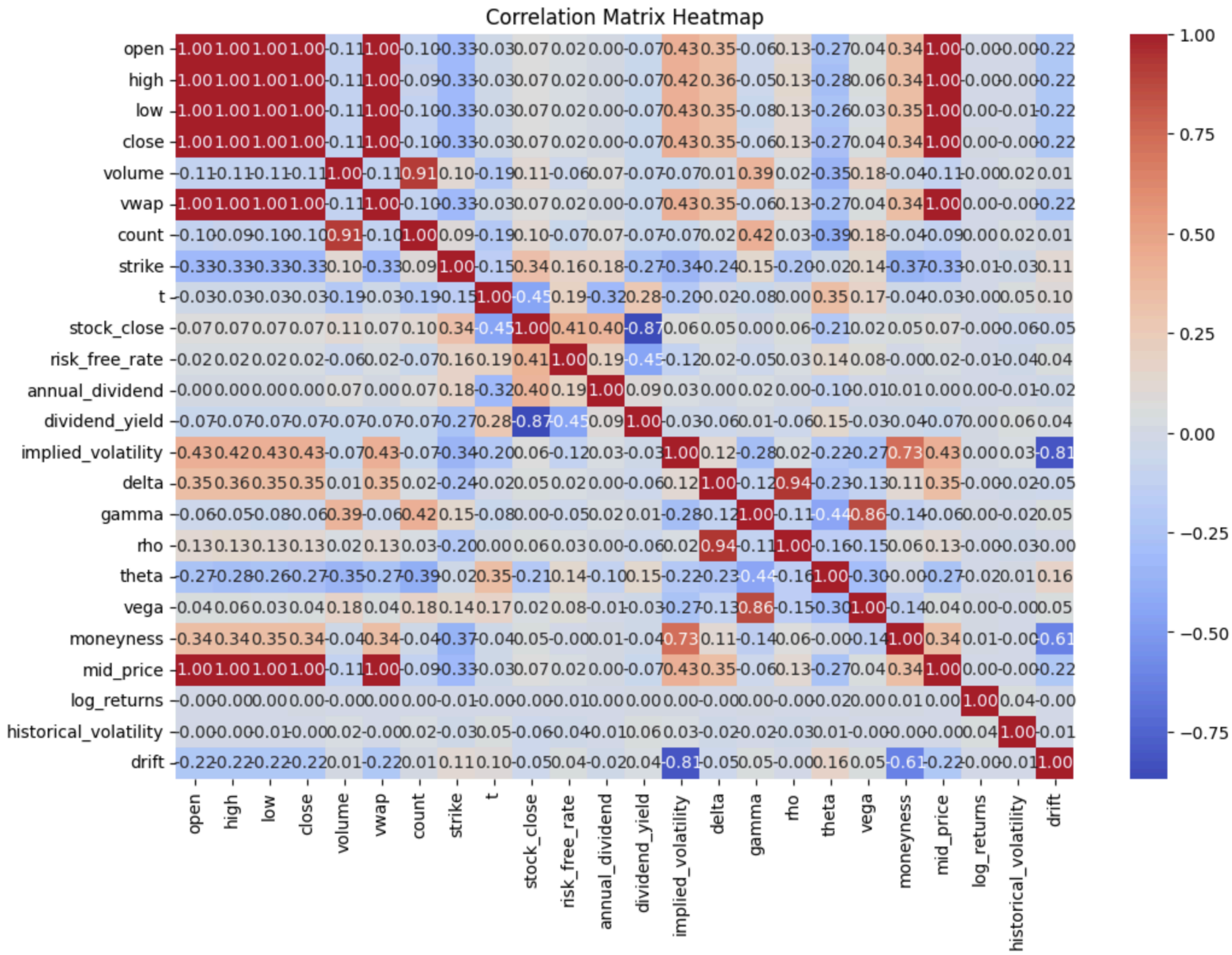

Figure 5: Feature Correlation Matrix

This heatmap shows how various features in the dataset are related to one another. We see that price-related variables like **Open**, **High**, **Low**, **Close**, and **VWAP** are almost perfectly correlated, meaning they provide redundant information and some can likely be removed from the model. **Implied volatility** is strongly tied to **vega** and **historical volatility**, which makes sense since vega measures an option's sensitivity to changes in volatility. Overall, by identifying these correlations, we can simplify the dataset by removing features that don't add unique value.

4. Methodology

As we have mentioned, we are using options trading data to infer whether the price of a stock will increase or decrease. Specifically, we are interested in predicting whether a stock's price will increase within a two-month window. However, it is important to note that we are dealing with various options contracts, including calls and puts, with different strike prices. Additionally, these contracts expire on different dates, and for each contract, we have extensive daily trading information leading up to their expiration.

Given this complexity, it is crucial to design a systematic approach to manage these large volumes of data, enabling us to structure it effectively across multiple tickers and apply it to modeling tabular data.

As a result, we have developed the following methodology to structure the entire project from end to end, from data gathering to real-time implementation. Below is the step-by-step methodology:

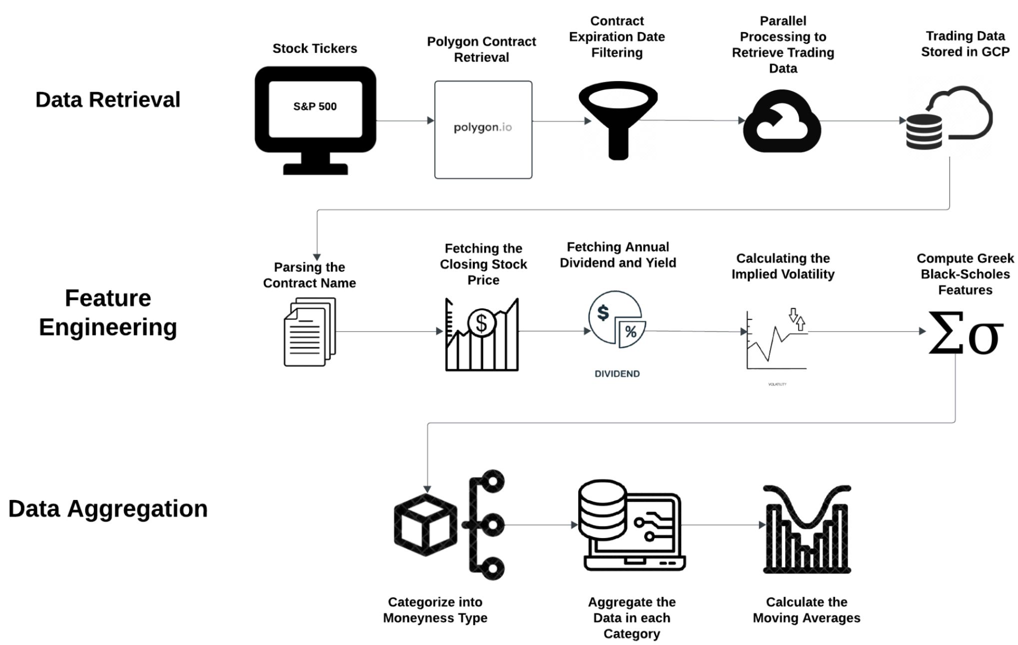

Figure 6: Methodology Pipeline

4.1. Data Gathering

We used financial data, including stock prices and options trading information, for our project. To obtain this data, we use Polygon. Specifically, we start by gathering trading and options data for all companies that belong to the S&P 500. The first is to gather the list of our S&P 500 stock tickers then retrieve the names of the put and call option contracts.

The process for collecting the data is as follows: We retrieve daily stock price data for all companies in the S&P 500, which includes the closing price. Additionally, we collect all the different option contracts that have expired for these companies over the last two years.

Once we have all the expired contracts, the tricky part begins: we need to gather the trading information for those contracts and determine the specific days for which we are collecting that data and for which contracts. To achieve this, we follow this approach: for each day x between 2022 and 2024, we find the contract that expires closest to $x + 60$ days, and then we gather all the trading data between x and the contract's expiration date. To collect the trading data we utilize GCP parallel processing to increase the number of workers to speed up the data extraction process.

As a result, for each day x, we have trading information for all the contracts (both puts and calls) with various strike prices that expire closest to $x + 60$ days. These contracts typically share the same expiration date, which is usually the third Friday of each month. Additionally, we filter out contracts that do not involve monthly expiration dates. Finally the data we retrieve was stored on the GCP storage.

4.2. Target Selection

As outlined in the project, we aim to build a model that predicts stock price movement over a 60-day (two-month) window. Therefore, our target variable is based on the change in the stock price two months into the future compared to the current day's closing price (specifically, the closing price of the previous day). Ultimately, our goal is to provide the user with a recommendation on whether the current stock is a buy, sell, or hold.

As a result, instead of approaching this problem from a regression perspective, we will categorize the price movement into three classes:

- The percentage increase is above a certain threshold.
- The percentage decrease is below a certain threshold.
- The stock's percentage movement remains between the defined thresholds (e.g., -x% to x%).

4.3. Feature Engineering

Once we’ve collected all the trading data for various options contracts—including details like strike prices and contract types—we move on to feature engineering to extract meaningful insights. The main features we’re working with include Open, High, Low, Close, Volume, VWAP (Volume-Weighted Average Price), and Count. We also bring in the risk-free rate and use these features, along with the Black-Scholes model, to calculate additional parameters that serve as inputs for our models.

The feature engineering process, shown in Figure 6, begins with parsing the contract name to extract important details like the ticker name, strike price, and expiration date. From there, we pull in additional data such as the closing price, annual dividend or dividend yield, and implied volatility. These variables allow us to calculate key Black-Scholes "Greek" values, such as delta and gamma. For instance, delta helps us understand how an option’s price changes relative to movements in the underlying asset, while gamma measures how delta itself changes in response to those price movements. These metrics give us valuable insights into the behavior of options prices and enhance the predictive power of our models.

4.4. Data Aggregation

Now, the only remaining problem is that we have a lot of trading information for option contracts for each day, but we need to find a way to ensure that we aggregate all the information by day. Why is that? We ultimately want to inform users on the current day whether a particular stock is a buy, sell, or hold, using all the option data available from the previous day to infer that decision. As a result, we need to aggregate the data so that it can fit into the model as a single observation for that particular day.

The way we do this is as follows: we must keep in mind that our approach may need to be adjusted if we encounter issues with prediction accuracy, as this part of the pipeline is crucial and strongly affects subsequent processes.

Here’s the approach: given a specific day, we use the closing price of the previous day and categorize the different options contracts into four distinct groups. We want to categorize the contracts because each group potentially provides different information, as follows:

- **Call options** with a strike price **above** the current stock price – information about how much the market believes the stock price will increase.
- **Call options** with a strike price **below** the current stock price – information about how much the market believes the stock price will not increase.

- **Put options** with a strike price **above** the current stock price – information about how much the market believes the stock price will not decrease.
- **Put options** with a strike price **below** the current stock price – information about how much the market believes the stock price will decrease.

By organizing the data into these four categories, we capture different types of information, each with its parameters, that reflect the market sentiment on stock price movements within a two-month window.

As for how we are going to aggregate this information, we used a weighted average based on the trading volume of each contract. There are many different contracts each day with various strike prices; some are rarely traded, while others are traded heavily. Therefore, using a weighted average by volume allowed us to give more importance to more actively traded contracts and their respective Black-Scholes features.

Lastly, we apply exponential moving averages (EMAs) to the data, which helps smooth out short-term fluctuations and highlight longer-term trends. This technique assigns greater weight to more recent data points, making it particularly useful for capturing momentum and identifying patterns in price movements. By incorporating EMAs, we can better understand market trends and make more informed predictions based on the underlying data.

Once we aggregated all the data by day, divided it into four groups, and applied a weighted average for each group's Black-Scholes features, we moved on to machine learning development. At that point, each observation will represent a specific day, with all trading information properly aggregated.

4.5. ML Modeling

Once we have all the data and have determined our target variable, we are ready to move on to the modeling phase of the project. We included XGBoost and LSTM in our experiments. XGBoost was tested both with and without Principal Component Analysis (PCA) for dimensionality reduction (reducing dimensions from 1500 to 200). Similarly, we applied PCA to LSTM and compared its performance with the non-PCA version of this time series model (this time the number of features was reduced from 50 to 22). Interestingly, we observed that all models, including XGBoost and LSTM, performed worse when PCA was applied, likely due to the loss of important information during dimensionality reduction.

Exponential Moving Average (EMA) was used for our XGBoost models. EMA added many features to the dataset because they are calculated over different time periods, such as 10, 20, or 50 days. Each time period captures trends at different granularities—short-term, medium-term, and long-term. These features provide valuable insights into market trends, momentum, and relationships between different metrics, enhancing the dataset and improving the predictive power of our models.

An important point to highlight is that for our LSTM, we avoided using a moving average to smooth the data, as it reduced noise in the time series. EMAs don't work well with LSTMs because they smooth out data and reduce short-term fluctuations, which can remove important patterns that LSTMs are designed to capture. LSTMs excel at learning sequential dependencies and relationships over time, so overly smoothed data from EMAs might limit their ability to rapid changes in the dataset.

While we segment the data into training and testing sets based on the observation date to prevent data leakage, the date itself is not a significant feature in the model. Instead, we treat each observation independently, as the moving average already accounts for how the data has been evolving leading up to the current day. This aspect will be a key focus of the project moving forward.

4.6. Evaluation

To begin the evaluation phase, we started with a basic train/test split using a small subset of the data to identify which models performed best. We initially tested Random Forest, XGBoost, and LSTM. After evaluating the results, XGBoost and LSTM stood out as the top-performing models. With these two models, we moved on to testing them on the complete dataset, which consists of data from S&P 500 companies. To evaluate their effectiveness, we ran simulations comparing the returns of XGBoost and LSTM, both with and without PCA, against the S&P 500 benchmark.

Our simulation process is illustrated in Figure 7. The pipeline begins by using six months of S&P 500 data for training, followed by predicting stock performance for the next two months. This process repeats, adding the next month's data to the training set and predicting the two months after that. At each step, the models identify the top five stocks most likely to increase or decrease in value based on the probabilities generated by XGBoost and LSTM. Depending on the predictions, $1,000 is either invested in each stock (if the stock is expected to rise) or shorted (if it's expected to fall). The models are retrained monthly using this expanding dataset.

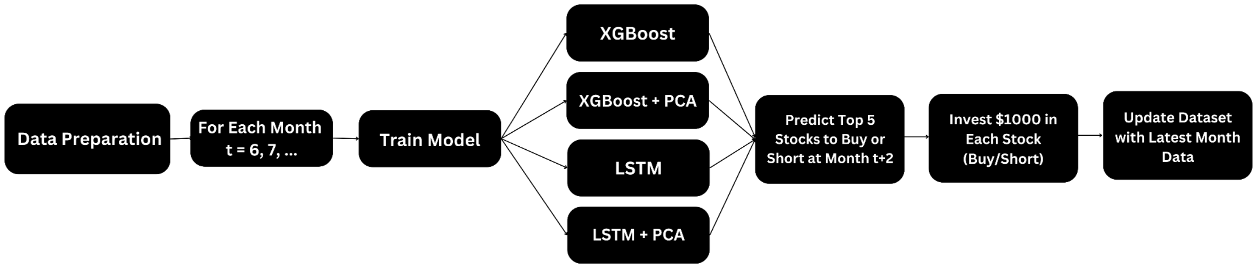

Figure 7: Simulation Pipeline

The results of the simulation are presented in Figure 8. Overall, XGBoost outperformed LSTM in most scenarios, with the best results achieved when PCA was not used to reduce dimensionality. However, further analysis is needed to determine if XGBoost consistently beats the S&P 500 benchmark. While some investments resulted in losses, the successful predictions generated substantial returns, significantly boosting overall ROI.

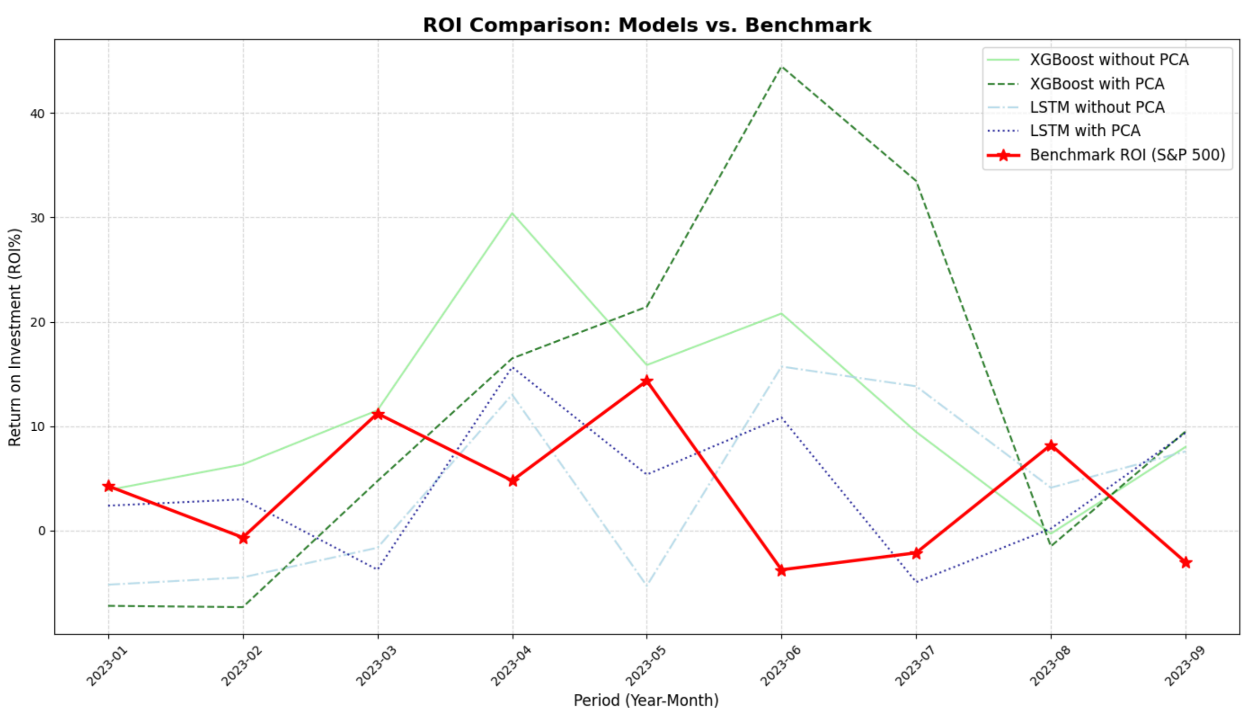

Figure 8: Simulation Results

5. Conclusion

Our project demonstrates the potential of leveraging options trading data to predict stock price movements and enhance investment strategies. By analyzing patterns such as unusual trading volumes or shifts in implied volatility, we were able to capture signals from informed traders that hint at future price changes. These insights, as validated by our simulations, show that options data can serve as a valuable supplement to traditional stock data for medium-term forecasting.

However, while the results are promising, there are important ethical considerations to address. The use of such models must ensure compliance with regulatory frameworks to avoid misuse of privileged or non-public information. Additionally, maintaining transparency in the model's outputs is crucial for building trust with stakeholders and ensuring ethical use. Care should also be taken to prevent strategies that could destabilize markets or disproportionately disadvantage certain participants, such as retail investors.

Overall, this project provides a strong foundation for creating robust, transparent, and responsible trading models that add value for market participants while upholding fairness and stability.

5.1. Future Work

To build on the findings of this project, there are several areas for further development:

1. **Expand the Dataset**
   Extending the dataset to include a longer historical time range and incorporating additional tickers can improve the model's generalizability and robustness. This would provide more comprehensive insights into market trends and help refine predictions across a wider range of stocks.
2. **Interactive Platform**
   Developing an interactive platform to make the insights generated by the model accessible and actionable for users. The platform would include clear visualizations and explanatory features, allowing users to easily understand and apply the predictions.
3. **Experimentation with Advanced Models**
   Exploring additional machine learning models, such as deep learning architectures like transformers, could uncover complex relationships in options data that traditional models might miss. These advanced models could help further improve prediction accuracy

By addressing these areas, future iterations of this work can aim to create a more comprehensive, user-friendly, and ethically responsible solution for leveraging options data in financial forecasting.

Appendix: Key Terms

- N: Cumulative distribution function (CDF) of the standard normal distribution.
- $N'(d_1)$: Probability density function (PDF) of the standard normal distribution.
- d_1 and d_2: Intermediary variables, calculated as:

$$d_1 = \frac{\ln\left(\frac{S_0}{X}\right) + \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

Where S_0 is the current stock price, X is the strike price, T is the time to expiration, r is the risk-free rate, and σ is the volatility.

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